



PROJECT REPORT

DATASET ANALYSIS, MACHINE LEARNING AND VISUALISATIONS

Data Science for Business Development, Fall 2025

Date of submission: 1. January 2026

Number of characters with spaces (excluding appendix): 63.734

Ruyi Zhong ruzho25@student.sdu.dk

Sasja Stein saste25@student.sdu.dk

Svetlana Podolskaja svpod25@student.sdu.dk

Lecturer

Prof. Jonas Husum Dalstrup jdals@sam.sdu.dk

Table of Contents

1. SQL and Database Analysis.....	4
1.1 Setting up the tables	4
1.2 SQL Queries	5
2. Data Preparation	6
3. Correlation & PCA	12
3.1 Grouped PCA.....	14
3.1.1 Customer Spend Behaviour	15
3.1.2 Customer Interaction and Engagement	16
3.1.3 Service and Experience.....	17
3.1.4 Product Preference.....	18
4. Clustering analysis based on PCA.....	19
4.1 Clustering Segments.....	22
Cluster 1: Highly engaged and high-value customers.....	22
Cluster 2: Inactive or disengaged customers	22
Cluster 3: At-risk customers with service issues.....	23
Cluster 4: Satisfied but low-engagement customers.....	23
5. Prediction Goals	23
5.1 Regression.....	23
5.2 Classification	24
5.3 Additional Supervised regression – <i>KNN with Customer Satisfaction Score</i>	25
6. Predictive Modelling.....	25
6.1 Regression.....	25
6.2 Classification	29
6.3 Additional Supervised regression – <i>KNN with Customer Satisfaction Score</i>	33
7. Power BI Dashboard	35
7.1 Customer Overview – Interactive Dashboard.....	35
7.2 PCA Analysis and Customer Segmentation	36
7.3 Regression.....	37
7.4 Risk and Experience prediction.....	38
8. Conclusion	39

Literature	41
Attachments	41
List of Figures	41
Appendix.....	43
1. Tables	43
1.1 customers.....	43
1.2 products	43
1.3 orders	43
1.4 order_items.....	43
1.5 order_returns	44
1.6 interactions.....	44
1.7 deliveries	44
2. Individual Questions	45
Ruyi Zhong.....	45
Sasja Stein.....	46
Svetlana Podolskaja.....	46
3. Cluster Scores	47

1. SQL and Database Analysis

1.1 Setting up the tables

Using the Schema Description from [*exam_database_description.pdf*](#) we created the following seven tables: customers, products, orders, order_items, returns, interactions and deliveries. For a more detailed overview of each table, see [Appendix – 1. Tables](#) in the bottom of our report, and the attached SQL file [*Tables_DataScienceExam.sql*](#).

We began by creating the header tables (customers, products and orders), by identifying appropriate data types and adding basic integrity rules, like non-negative numeric values and relevant constraints. As we progressed, we introduced foreign keys to map our tables together and ensure meaningful relationships between customers and their orders.

Overall, the tables were composed in a such way that: one customer can place many orders, but each order belongs to exactly one customer; each order can contain many order items; each order item references exactly one product, while a product can appear in many order items across different orders; a customer can have many interactions, and each interaction belongs to exactly one customer. Additionally, deliveries are stored at order level with one delivery per order, while returns are linked to orders and products. Finally, we added a couple of CHECK constraints to limit invalid inputs and ensure data integrity, accuracy, and consistency.

In addition to the overall structure, several design decisions were made at table level to further improve data quality and usability. These included setting realistic limits on customer birth years, standardizing categorical fields such as region, gender, loyalty tier, and payment method, and ensuring that prices, quantities, and order values are non-negative while still allowing zeros for realistic cases like free products or full discounts. We also aimed to keep the schema clear and minimal while avoiding technical conflicts. Therefore, composite primary keys were used in the order_items table to prevent duplicate order lines and to represent multiple quantities of the same product efficiently. Furthermore, we have renamed the returns table to order_returns to avoid conflicts with SQL reserved keywords. Next, some channel-specific fields in the interactions table were allowed to be NULL when not applicable. Last, but not least, deliveries in the delivery table were modelled with a one-to-one relationship to orders by using order_id as the primary key, ensuring simplicity while accurately reflecting the underlying business logic.

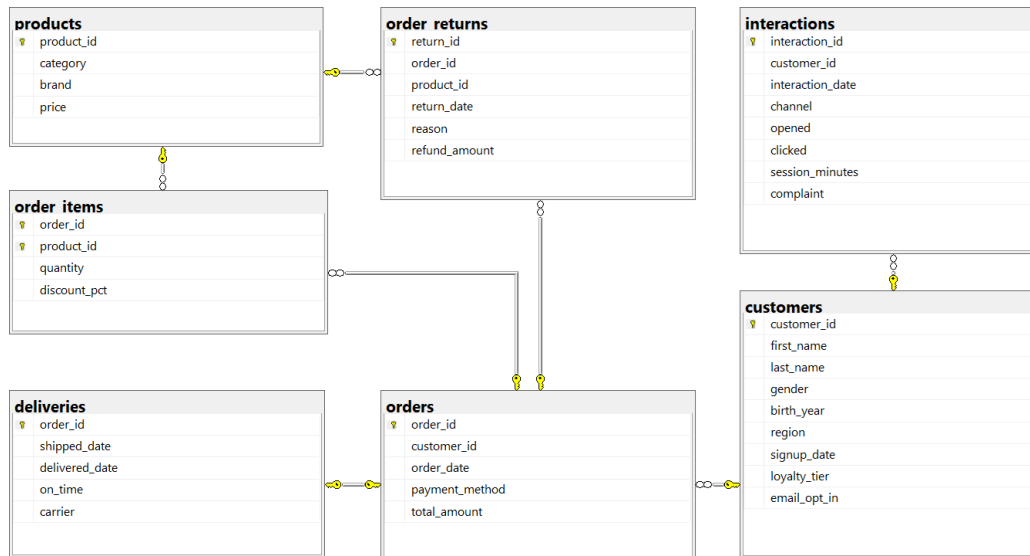


Figure 1 - Database overview in SQL

1.2 SQL Queries

A total of twenty-four queries were designed to help have a deeper understanding of the data given. Eight queries were assigned evenly to each group members. For further details on each individual question, see [Appendix – 2. Individual Questions](#). Furthermore, our SQL queries can be found in the attached file [Queries_DataScienceExam.sql](#)

First of eight queries were given by Ruyi Zhong. Among these eight queries, first two queries were surrounding customer segmentation and get basic understanding of who are the business loyal customer group. The following queries (3 to 6) were designed to understand the products, including the most profitable products, most popular products and such, thus we can get a full knowledge on which kind of products the business should focus on and have further proact marketing as well as pricing strategy. The last 2 queries are about service and complaints. From these 2 queries, we can know the customer satisfaction as well as the service level the business has been providing.

The next eight queries were developed by Sasja Stein and focus on customer value, purchasing behaviour, payment preferences, and regional performance. The first set of queries examines loyalty tier insights by comparing average order value across Premium, Plus, and Basic customers, allowing for an assessment of whether higher loyalty tiers are associated with increased spending. Product-related queries then explore overall product popularity as well as the most popular products within each loyalty tier, highlighting differences in purchasing behaviour between customer segments.

Payment method insights analyse revenue distribution across different payment methods and investigate whether payment preferences are associated with customer engagement, measured through email opt-in behaviour. Finally, regional and delivery-focused queries examine customer and order distribution by region, delayed deliveries, and the relationship between delivery delays and customer complaints. Together, these queries provide a comprehensive view of how geography and logistics influence customer experience and satisfaction.

The final eight queries were developed by Svetlana Podolskaja and focus on operational efficiency, marketing effectiveness, customer behaviour following service interactions, and return patterns. First queries analyse time-based relationships also between regions, especially delays and whether such delays are linked to higher return rates. The third and fourth queries assess the effectiveness of marketing channels by examining customer behaviour following promotional interactions, particularly among younger customer segments. Further queries investigate return behaviour in depth, including whether certain products are returned more frequently than they are kept, whether size-related returns dominate return reasons, and whether shipping speed influences return behaviour. Finally, repeated purchasing behaviour is analysed to identify customers who frequently order the same products in large quantities, which could indicate opportunities for personalized marketing, loyalty programs, or bulk-purchase incentives.

Overall, the set of queries covers customer segmentation, product performance, marketing effectiveness, service quality, logistics, and return behaviour. Together, these analyses provide a well-rounded understanding of customer behaviour and operational performance, forming a strong foundation for potential further analysis.

2. Data Preparation

After loading and carefully inspecting the given dataset, we analysed and prepared it for later manipulations, by following the steps described further in the report. All analytical steps described in this report were carried out in Python and can be found in the attached script *Data_Science_Exam.py*, which contains the full data preparation, analysis, and modelling workflow.

Overall, the original dataset contains 6000 rows and 37 columns. Let's look in our dataset a little bit more deeply to understand our priorities. First, we check the column names to identify those which we do not need later, so we can create a subset. One of the first things that we want to look after is numerical values. In our dataset only two columns, region and gender, have no numeric values (maybe it makes sense to convert them into numerical with encoding). Then we check them on missing values.

Out of 6000 rows in total there are 284 missing rows in `avg_order_value`, 309 in `returns_rate`, 1269 in `satisfaction_score`, 1675 in `email_open_rate`, 1458 in `email_click_rate`, which could be a case, since they are only relevant if the customer receives emails, and 314 in `future_3m_spend`. In total there are 5309 missing values through the whole subset. There are no duplicate rows or negative values. We call a `describe` function then for our numeric values, which shows us total, mean, min and max values as well as std.

For 284 missing rows in `avg_order_value`, we applied medium imputation. This choice was based on several considerations. First, `avg_order_value` is a stable, central tendency measure, making the median an appropriate replacement. Second, Median is robust against outliers, whereas mean could be skewed by high-value customers. In addition, the proportion of missing values was low (<5%), so simple imputation is appropriate. We also consider other imputation methods, such as k-nearest neighbors (KNN) or regression-based approaches. However, these methods will add unnecessary complexity for a variable with low missingness, and they may distort the original distribution more than a median imputation.

When checking the missing values in `avg_order_value`, we also discovered invalid values in the `total_orders` and `total_spent` columns. Although these fields did not contain missing values, there are 2,352 rows had `total_spent` = 0 and `total_orders` = 0 despite having positive order-level statistics (e.g., non-zero average, median, and maximum order values, as well as discount metrics). This inconsistency indicates that both totals were incorrectly recorded and should be treated as missing. Because both totals were missing simultaneously, formula-based imputation was impossible.

To address this, we trained a Random Forest Regressor to predict `total_spent` based on features that remain intact even when totals are missing. (such as `avg_order_value`, `median_order_value`, `max_order_value`, `discount_rate_mean`, `pct_orders_discounted`, etc.) We selected Random Forest for several reasons. First, we limit our Machine learning algorithm down to supervised continuous machine learning framework since `total_spent` is a continuous variable. And because spending behaviour is complex and unlikely to be linear, making linear regression unsuitable. In the end, we first try to use Random Forest, which maps the customers' behaviour and then predicts their spending.

For evaluation, we applied both 5-fold cross-validation (CV) and out-of-bag (OOB) validation. Both applied in 80% training set of valid data and 20% testing set of valid data. And we can see the results as following:

```

=== OOB Approach ===
OOB R2 (internal on train): 0.8790
Test R2 : 0.8711
Test MAE: 61.67
Test RMSE: 83.31

=== CV Approach (5-fold on TRAIN) ===
CV R2 (internal via OOF preds): 0.8767
CV MAE: 67.52
CV RMSE: 95.31
Test R2 : 0.8711
Test MAE: 61.67
Test RMSE: 83.31

=== Comparison Summary (TEST set) ===
OOB -> R2=0.8711, MAE=61.67, RMSE=83.31
CV -> R2=0.8711, MAE=61.67, RMSE=83.31

```

Figure 2 - Results of OOB and CV approaches for validation

These results show that both evaluation methods produce very similar performance estimates. This indicates that the model is reliable, stable, and generalizes well, and that both OOB and 5-fold CV are suitable for validation approaches for this case. We applied OOB to impute invalid data due to the slightly well performance.

However, when we look into the comparison between original data and the imputed values (as below picture 3), Random Forest has given a much higher predicted value. This result highlighted few things we have not considered before: First, Random Forest is optimized for predictive accuracy by estimating the conditional expectation of all given variables, rather than for preserving the distributional properties of original data. The result implied that customers with missing data are, on average, richer than those with fully observed data (Inflationary bias), while we do not have evidence to support that claim. Moreover, the missingness in total_spent and total_orders is not random. The corrupted rows represent approximately one third of the dataset and exhibit strong correlations with other variables, suggesting a Missing Not at Random (MNAR) mechanism. Under MNAR, customers with higher activity or value are more likely to have missing or corrupted totals. Therefore, although the Random Forest-imputed values are suitable for predictive modelling, they are not appropriate for financial reporting or descriptive economic analysis.

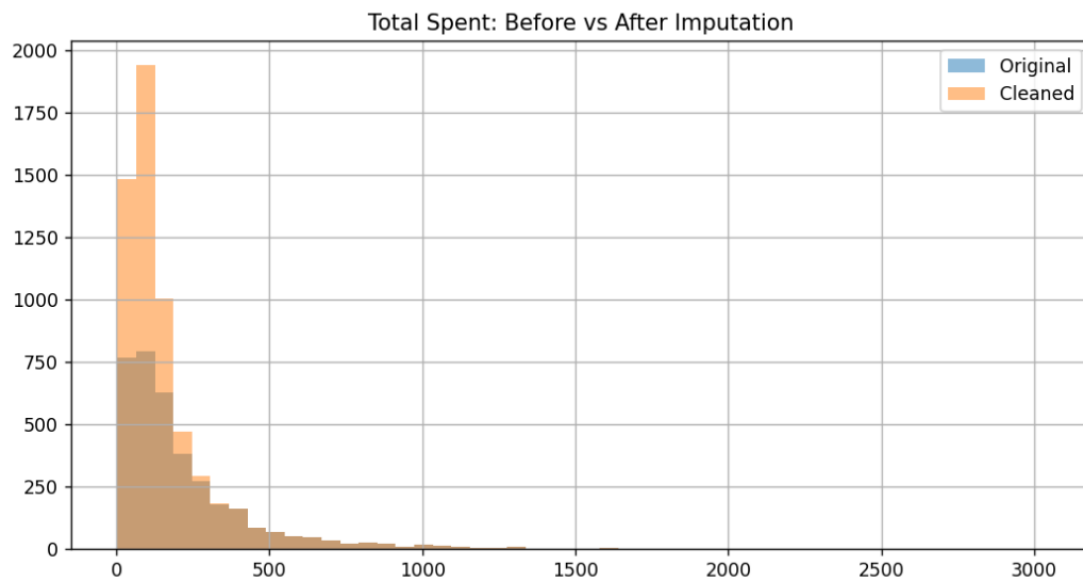


Figure 3 - Total spent: Before vs After Imputation

For exploratory and descriptive analyses, a KNN-based imputation was then used to preserve local similarity and population-level distributions. KNN approaches the missing data in a different way from Random Forest. While Random Forest estimates conditional expectations and can amplify Missing Not At Random (MNAR) mechanisms, KNN relies on local similarity and produces more conservative estimates. For example, when Random Forest sees ‘high engagements’ and predicts it as high spend, KNN imputes values based on the observed spending of comparable customers. As a result, KNN-imputed values tend to be lower and exhibit reduced variance compared to fully observed data. Picture 4 below shows the distribution of total spending before and after KNN imputation. We can see that KNN introduces a conservative bias, but it avoids artificial inflation of spending and prevents the creation of extreme synthetic values. The KNN-imputed total spending follows the lower and middle range of the observed distribution without introducing a new right tail. Therefore, KNN imputation results will be used for exploratory analysis, such as correlation heatmap, PCA and clustering, and are not applied in financial reporting.

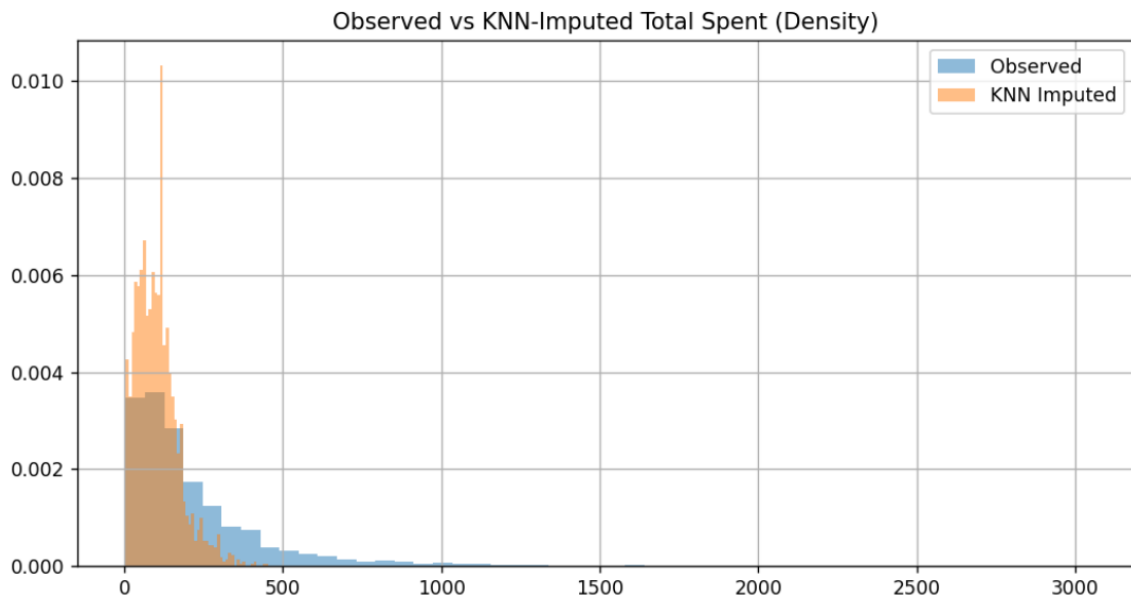


Figure 4 - Total spent: KNN imputation

After predicting total_spent for the corrupted rows, we derived the corresponding total_orders values using the relationship as following: $\text{total_orders_pred} = \text{total_spent_pred} / \text{avg_order_value}$.

For interactions via email we introduced new value 'NR = not related', since they are only related in case a customer signed up for receiving marketing emails. By this we can reduce the amount of missing data in rate for opened email (originally, 1675 rows missing) and rate for clicking the link in the email (1458). First, we needed to change the data type of these two columns to 'object', in order to allow inserting strings, not only numeric values. Secondly, we introduced the mask, which would retrieve only the necessary rows, changing the NaN = not a number to NR. By doing this, we managed to fill all the missing data in email click rate, leaving us with only 217 missing values in opened email rate. To impute the missing values in satisfaction score parameter, we used the median value, since it seemed reasonable with 1269 rows missing and the range of satisfaction was from 0 to 10.

Detection and removal of outliers is another important step in data preparation process, so these extreme values would not affect further calculations, like average value, or mislead our predicting models with invalid assumptions. There are several ways to find outliers, namely, Z-Score method, inter quartile range and percentile methods, however, in our case we decided to proceed with the simplest and fastest one – visualisation, specifically, by building histograms. For some of the columns this is not quite relevant, for example, customer id or whether the customer gets emails, but for the other parameters, such as forecasted spending in the next three months, the histogram illustrates the extremely high values, which we might need to remove. The columns with the most obvious and

suspicious outliers would be: loyalty points, number of total orders, total amount of money spent, days since first order, average delivery and future spendings in three months. Let's handle the last one. On the one hand, we could just delete those rows, but, on the other we can replace them as well as the missing values with either average or min/max. In order to do this we call the .describe function on that column, which gives us following information:

count	5686.000000
mean	28.937979
std	43.955245
min	0.000000
25%	0.000000
50%	4.690000
75%	46.267500
max	371.290000

In this case, we would assume that everything bigger than 75% tile value is too big, so we want to shrink it. We find all the rows with mask and then change outliers to the 46.267500. For missing data, we fill it with new mean (17.634899).

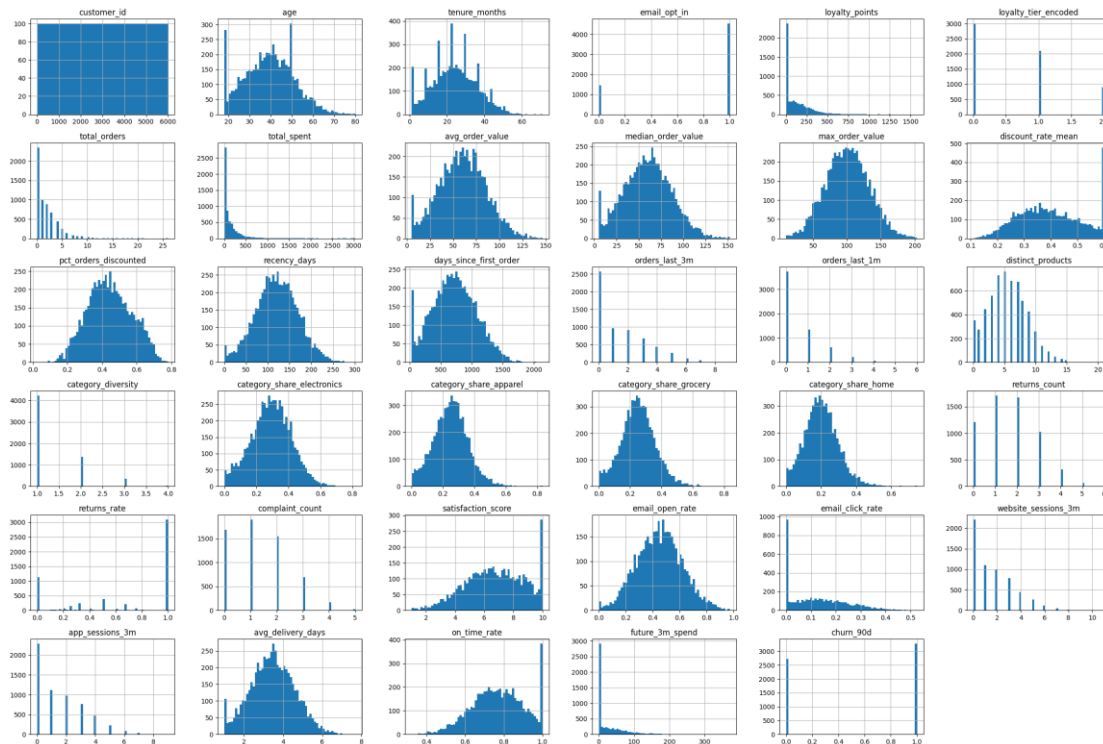


Figure 5 - Distribution of values by given parameters

3. Correlation & PCA

After cleaning dataset, the correlation heatmap was used to visualize linear relationships among numeric variables and also to identify the features that have overlapped information. As below heatmap (picture 6), it shows few pairs of variables with strong positive or negative correlations, which indicate the similarity of given information.

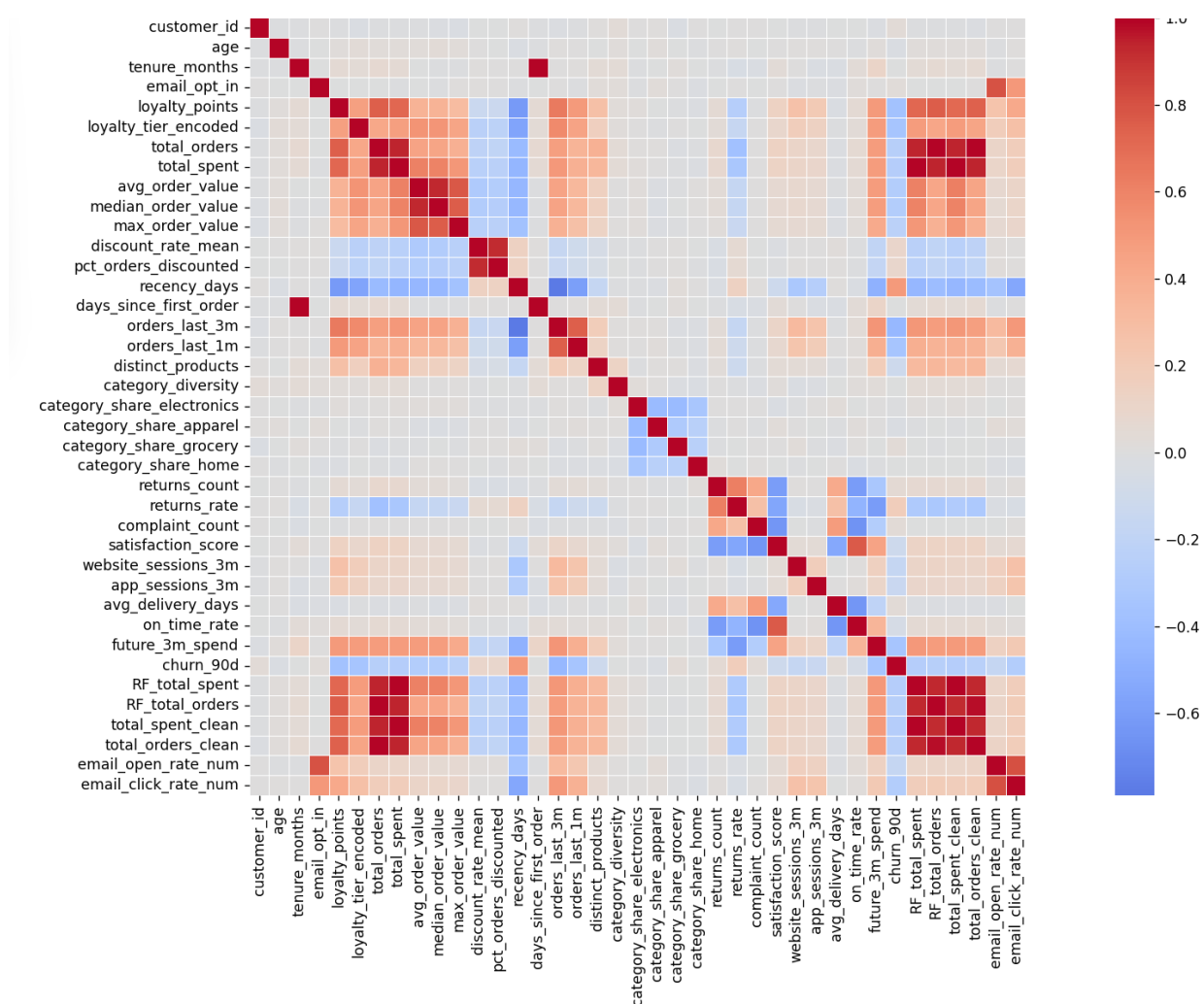


Figure 6 - Correlation Heatmap

Including all highly correlated variables would introduce redundancy, inflate variance in certain dimensions, and reduce the interpretability of the PCA results. Therefore, it is necessary to drop them before PCA. In addition, there are also some of the variables that are Boolean or categorical datatypes. Since PCA is designed for continuous numeric variables, not categorical or Boolean variables, it is also necessary to drop them before PCA. Furthermore, Predictive outputs, corrupted transactional totals, and future-looking variables were also excluded to prevent information leakage and circularity.

Based on above analysis, we have excluded the following variables: “customer id”, “region” and “gender” due to them being non-numeric identifiers; “encoded loyalty tier”; spend and order duplicates such as total spent and orders both actual and predicted and also predicted imputation “total_spend_clean”, “total_orders_clean”, “RF_total_spend” and “RF_total_orders”; median and maximum order value; “days since first order”; “orders per last month”; count of returns and category diversity, all to reduce noise; and, finally, future three months spendings and churn possibility as they have no leakage into PCA.

To explore the overall latent structure of customer behaviour, an initial Principal Component Analysis (PCA) was conducted using all available numeric variables after data cleaning and standardization. This is aimed at checking whether a small number of global components could adequately summarize customer behaviour.

=== ALL VARIABLES PCA – Explained Variance ===			
	Eigenvalue	Proportion of Variance	Cumulative proportion
PC1	4.3042	0.1956	0.1956
PC2	3.1122	0.1414	0.3370
PC3	2.0641	0.0938	0.4308
PC4	1.4850	0.0675	0.4983
PC5	1.2886	0.0586	0.5569
PC6	1.2327	0.0560	0.6129
PC7	1.1428	0.0519	0.6648
PC8	1.0014	0.0455	0.7103
PC9	0.9937	0.0452	0.7555
PC10	0.8782	0.0399	0.7954
PC11	0.8279	0.0376	0.8330
PC12	0.7743	0.0352	0.8682
PC13	0.6910	0.0314	0.8996
PC14	0.5249	0.0239	0.9235
PC15	0.4162	0.0189	0.9424
PC16	0.3074	0.0140	0.9564
PC17	0.2748	0.0125	0.9689
PC18	0.2164	0.0098	0.9787

Figure 7 - PCA performed on all variables

As above result from PCA on all numeric variables, we can see the first principal component (PC1) explains approximately 43.1% of the total variance and more than ten components are required to exceed 80% cumulative explained variance. This suggests that that variance is widely dispersed across dimensions and thus customer behaviour in this dataset is multi-dimensional and cannot be effectively summarized by only a small number of global latent factors.

In detail, we can see the variables contributing proportion in PCA 1 to PCA 3 as below:

	PC1	PC2	PC3
age	-1.328126e-02	-5.847217e-03	-1.986479e-02
tenure_months	-8.537174e-03	3.544171e-02	1.511892e-02
email_opt_in	-2.220446e-16	-2.775558e-17	2.081668e-16
loyalty_points	-3.685349e-01	-6.648949e-02	-9.650738e-03
avg_order_value	-2.308318e-01	-3.776066e-02	-2.793677e-01
discount_rate_mean	1.012190e-01	2.065699e-02	6.354710e-01
pct_orders_discounted	9.374510e-02	2.083372e-02	6.336128e-01
recency_days	4.198265e-01	1.168270e-01	-2.033627e-02
orders_last_3m	-4.107366e-01	-1.055383e-01	-4.707643e-03
distinct_products	-1.157660e-01	7.066165e-03	-6.444821e-02
category_share_electronics	-1.792316e-02	-3.246511e-02	-4.074862e-02
category_share_apparel	-9.948463e-03	2.268580e-02	4.291596e-02
category_share_grocery	1.730308e-02	8.371648e-03	1.961848e-02
category_share_home	1.487026e-02	7.316826e-03	-1.751436e-02
returns_rate	1.630365e-01	-3.073000e-01	4.628194e-02
complaint_count	7.857641e-02	-4.383137e-01	-2.750643e-02
satisfaction_score	-1.584381e-01	4.712432e-01	2.653795e-02
website_sessions_3m	-2.103332e-01	-7.422716e-02	1.181861e-01
app_sessions_3m	-1.765144e-01	-5.623565e-02	1.063841e-01
avg_delivery_days	8.152786e-02	-4.123871e-01	-1.030586e-02
on_time_rate	-1.151614e-01	4.956186e-01	2.427065e-02
email_open_rate_num	-3.643374e-01	-1.188368e-01	2.000066e-01
email_click_rate_num	-3.826340e-01	-1.354828e-01	1.938539e-01

Figure 8 - PCA performed on all variables, continued

The first three principal components capture distinct aspects of customer behaviour. PC1 is driven by strong positive loading on recency_days (0.42) and strong negative loading on engagement variables. This component captures a recency–engagement dimension. PC2 reflects service quality and customer experiences. It states strong positive strong positive loading on on_time_rate (0.50) and satisfaction_score (0.47), and strong negative loading on avg_delivery_days (–0.41) and complaint_count (–0.44). PC3 is driven more by discount_rate_mean (0.64), which represents a price sensitivity dimension.

Other variables have a more limited influence on the leading components but still provide additional behavioural insights. For example, PC4 isolates product diversity behaviour, PC5 captures customer satisfaction and complaint patterns, and PC11 reflects return-prone behaviour. The remaining components explain relatively small proportions of variance and primarily represent noise or highly specific behavioural patterns.

3.1 Grouped PCA

PCA assumes that input variables measure a common underlying phenomenon. However, from the above result of overall variable PCA analysis, customer behaviour spans multiple distinct dimensions, such as spending, engagement, service quality, and product preference that are only loosely related.

Therefore, instead of applying a single PCA to all variables, we separate PCA models conducted different groups of variables representing distinct behavioural constructs. To preserve conceptual clarity, variables were grouped into customer spend behaviour, customer interaction and engagement, service and experience, and product preference. These 4 groups also are helpful for giving directions to further analysis and prediction.

3.1.1 Customer Spend Behaviour

The first PCA applied to customer spend behaviour variables revealed a clear low-dimensional structure. The first three principal components explained around 84% of the total variance which indicates that customer spending behaviour can be effectively summarized using these three dimensions.

```

=== Customer Spend Behaviour PCA - Explained Variance ===
  PC  Eigenvalue  Variance_Ratio  Cumulative
0  PC1   2.296538    0.459231    0.459231
1  PC2   0.985124    0.196992    0.656223
2  PC3   0.931884    0.186346    0.842569
3  PC4   0.576267    0.115234    0.957803
4  PC5   0.211020    0.042197    1.000000

Top contributors to Customer Spend Behaviour PCA PC1:
orders_last_3m      0.574969
recency_days       -0.571771
avg_order_value     0.480531
discount_rate_mean -0.252856
distinct_products   0.218273
Name: PC1, dtype: float64

Top contributors to Customer Spend Behaviour PCA PC2:
discount_rate_mean  0.860388
distinct_products   0.273418
avg_order_value     -0.255391
recency_days        -0.245317
orders_last_3m      0.244071

```

Figure 9 - Customer Spend Behaviour, PCA results

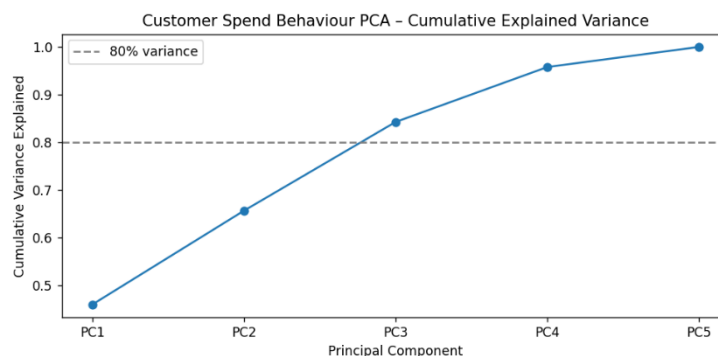


Figure 10 - Customer Spend Behaviour, Variance

The first principal component was mainly driven by order frequency, average order value, and recency and represents overall customer value and engagement. Customers with high scores on this component tend to purchase more frequently, spend more per order, and have purchased recently. The second component was highly related to discount usage, indicating differences in price sensitivity among customers. The third component reflected different spending styles, which separating customers who place fewer large orders from those who make more frequent but smaller purchases.

3.1.2 Customer Interaction and Engagement

The interaction and engagement PCA showed an effective dimensionality reduction, with the first three components explaining over 80% of the total variance. The first component was driven primarily by email open rates, click rates, and loyalty points, representing overall engagement intensity. Customers with high scores on this component interact frequently with marketing communications and loyalty programs.

```

=== Customer Interaction / Engagement PCA - Explained Variance ===
  PC  Eigenvalue  Variance_Ratio  Cumulative
0  PC1    2.484961      0.496909      0.496909
1  PC2    0.861287      0.172229      0.669138
2  PC3    0.805915      0.161156      0.830294
3  PC4    0.635282      0.127035      0.957330
4  PC5    0.213388      0.042670      1.000000

Top contributors to Customer Interaction / Engagement PCA PC1:
email_click_rate    0.556434
email_open_rate     0.540172
loyalty_points      0.440664
website_sessions_3m 0.341496
app_sessions_3m     0.296296
Name: PC1, dtype: float64

Top contributors to Customer Interaction / Engagement PCA PC2:
app_sessions_3m     0.925153
email_open_rate     -0.242208
email_click_rate     -0.200152
loyalty_points      -0.170918
website_sessions_3m 0.127099
Name: PC2, dtype: float64

```

Figure 11 - Customer Interaction and Engagement, PCA results

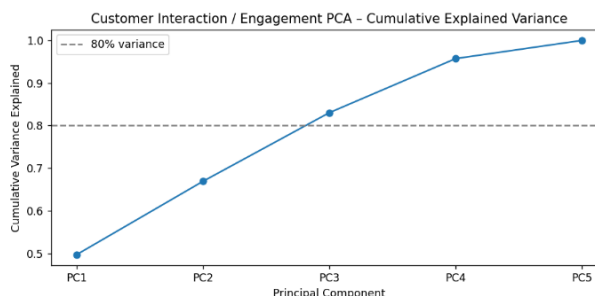


Figure 12 - Customer Interaction and Engagement, Variance

The second component highlighted differences in engagement channels, particularly contrasting app usage with email interaction. This suggests that customers differ not only in how engaged they are, but also in how they prefer to interact with the company. These components offer a structured way to understand customer engagement beyond raw activity counts and give a direction for further customer segmentation analysis.

3.1.3 Service and Experience

The PCA based on service and experience variables showed a particularly strong dominant component, with the first principal component alone explaining over 60% of the total variance. This component was positively related to on-time delivery and satisfaction scores, while it is negatively related to complaint counts, return rates, and delivery delays. Overall, it represents an overall service quality and customer experience dimension, where higher scores indicate smoother delivery performance and better perceived service outcomes.

```

=== Service & Experience PCA - Explained Variance ===
  PC  Eigenvalue  Variance_Ratio  Cumulative
0  PC1    3.201259      0.640145    0.640145
1  PC2    0.785003      0.156974    0.797120
2  PC3    0.527130      0.105408    0.902528
3  PC4    0.270170      0.054025    0.956553
4  PC5    0.217272      0.043447    1.000000

Top contributors to Service & Experience PCA PC1:
on_time_rate      0.506923
satisfaction_score 0.501169
complaint_count   -0.436699
avg_delivery_days -0.416628
returns_rate      -0.357174
Name: PC1, dtype: float64

Top contributors to Service & Experience PCA PC2:
returns_rate      0.829247
avg_delivery_days -0.384225
complaint_count   -0.375508
satisfaction_score -0.133417
on_time_rate      0.076909

```

Figure 13 - Service and Experience, PCA detailed

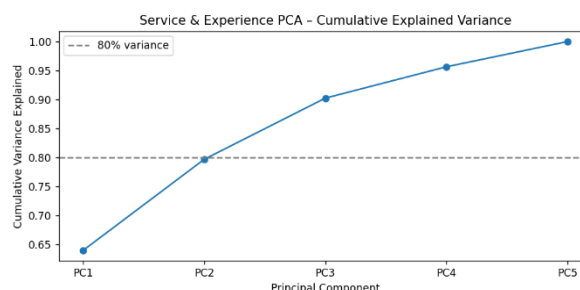


Figure 14 - Service and Experience, Variance

The second component captured operational friction, contrasting return behaviour, and delivery delays with other experience measures. This suggests that customer service experience is primarily shaped by overall service quality, with secondary differences driven by operational issues.

3.1.4 Product Preference

For product preference variables, the first two principal components explained nearly 90% of the total variance. The first component separates customers from those mainly purchase electronics and from those purchase more groceries and home categories. The second component further distinguished between home and grocery preferences. Since these components capture a large share of the variance, the results suggest that customers can be segmented by their product preferences, especially between electronics-focused and daily-need-focused customers.

```

=== Product Preference PCA - Explained Variance ===
   PC  Eigenvalue  Variance_Ratio  Cumulative
0  PC1    1.439231         0.479664         0.479664
1  PC2    1.237734         0.412509         0.892173
2  PC3    0.323536         0.107827         1.000000

Top contributors to Product Preference PCA PC1:
category_share_electronics    0.764951
category_share_grocery       -0.591222
category_share_home          -0.255551
Name: PC1, dtype: float64

Top contributors to Product Preference PCA PC2:
category_share_home           0.812672
category_share_grocery       -0.559955
category_share_electronics   -0.161290
Name: PC2, dtype: float64

```

Figure 15 - Product Preference, PCA results

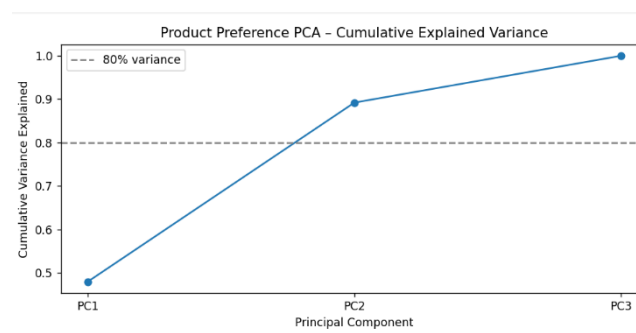


Figure 16 - Product Preference, Variance

In summary, the result of initial PCA on all variables shows that customer behaviour is highly multi-dimensional and cannot be adequately summarized using a small number of components. By using grouped PCA models instead, the analysis can achieve a greater interpretability while preserving most

of the relevant variance within each behavioural domain. The results from grouped PCA demonstrate that customer behaviour can be effectively summarized using a small number of meaningful latent dimensions. These components capture the dominant patterns in spending, engagement, service experience, and product preference.

Furthermore, the extracted components can be used into predictive modelling tasks, such as churn prediction or future spending estimation. Because the principal components are independent from each other, they reduce strong overlap between variables and help produce more stable model results. As a result, PCA not only enhances interpretability but also supports the development of more robust predictive models.

4. Clustering analysis based on PCA

Clustering, as an unsupervised classification algorithm, was used for underlying customer segments without predefined labels. By segmenting customers, we can identify business targets more directly and precisely and thus provide more relevant services, prices, and products to increase value proposition for customers.

Although clustering can be performed directly on the original variables, it may lead to distorted distance calculations when variables are highly correlated or noisy. In this context, the previously conducted grouped PCA can be naturally applied to support the clustering analysis as it was designed around key customer behaviour dimensions. By using grouped PCA scores for clustering, we can reduce the redundant and high correlated information and thus distance calculations become more reliable. Furthermore, the principal components also show a clear behavioural dimension, such as spending patterns, engagement level, service experience, and product preferences, which provide a solid foundation for customer segmentation.

To ensure robust customer segmentation, both K-means clustering and hierarchical clustering were applied. Although both methods belong to hard clustering, they represent different clustering philosophies: K-means is a partitional, prototype-based algorithm, while hierarchical clustering is a structure-based approach that builds a nested grouping of observations. Using both methods allows cross-validation of the segmentation results and reduces the risk that the identified clusters are an artifact of a single algorithm.

K-clustering was first applied. When it comes to the decision of K numbers, we have used the Elbow method and the Silhouette score. As the figures below, we can see that the Elbow curve shows a large reduction when increasing K from 2 to 3, followed by a moderate improvement from 3 to 4. After K=4,

the curve gains become marginal. The Silhouette score reaches its maximum at K=2, but drops sharply for K=3–4 and then stabilizes for larger values of K. Both methods shows that K=2 provides the strongest numerical separation, however, it will lead to overly broad segments with limited business meaning. Therefore, K=4 was selected in the end. Cluster sizes further support this choice.

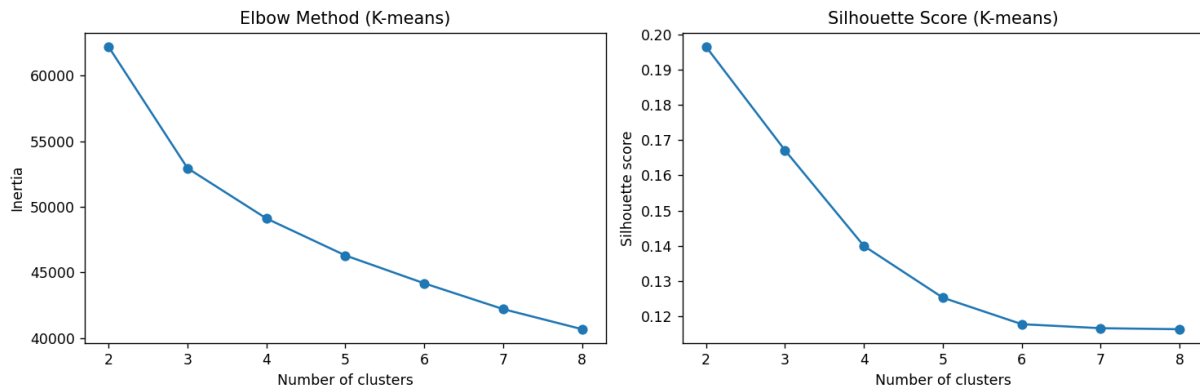


Figure 17 - K-means clustering

Hierarchical clustering with Ward linkage was then applied to the same PCA-based feature space. Ward's method merges clusters by minimizing the increase in SSE and is conceptually aligned with the K-means objective. Because hierarchical clustering does not require the number of clusters to be specified before starting, there is no further discussion of choosing K amount. Instead, in hierarchical clustering, the algorithm initially treats each observation as its own cluster and then repeatedly merges clusters until all observations belong to a single cluster. Therefore, it is important to decide where to stop the merging process based on the dendrogram. As below dendrogram, cutting the dendrogram at four clusters preserves meaningful customer groupings before such large distance increases occur.

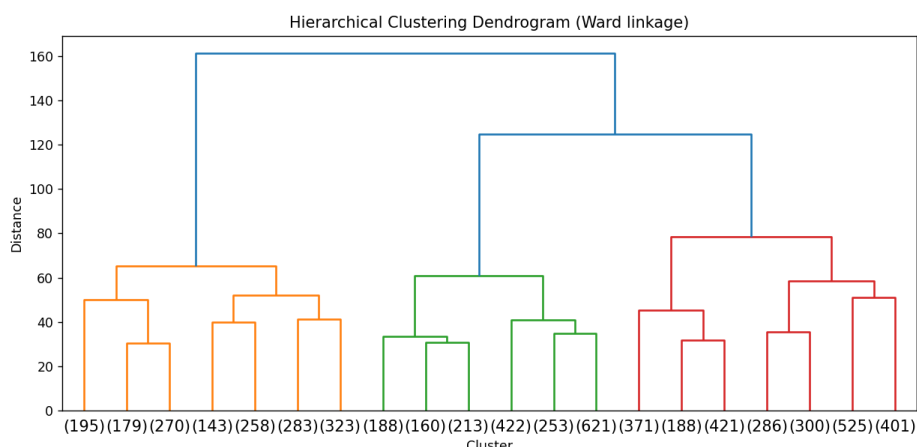


Figure 18 - Clustering Dendrogram

From the results of both K-mean and hierarchical clustering below, we can see that even though both clustering was decided to use 4 clusters, the resulting cluster assignments differ due to the different

clustering mechanisms. As shown in the K-means visualization, the clusters appear more balanced and better aligned with the main behavioural directions captured by the principal components. In contrast, hierarchical clustering is more sensitive to early merging decisions in the data. This leads to clusters that are less evenly distributed and show greater overlap, especially in regions where customer behaviors are more similar. Therefore, K-means clustering is more suitable for this specific analysis and used for this analysis.

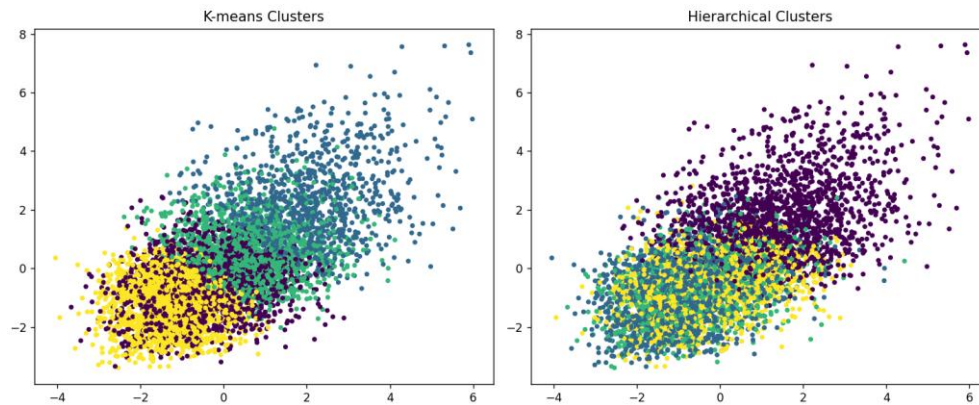


Figure 19 - Comparison of K-means and hierarchical clusters

To visually validate the clustering results, K-means clusters were plotted in PCA space using Spend PC1 and Engagement PC1. Each point in the plot corresponds to an individual customer, and the color indicates the cluster to which the customer was assigned by the K-means algorithm. Although clustering was performed using multiple components, projecting onto these two key dimensions allows inspection of cluster separation. The plot shows distinct customer groups with different combinations of spending and engagement, confirming that the clustering captures systematic behavioural patterns rather than random variation.

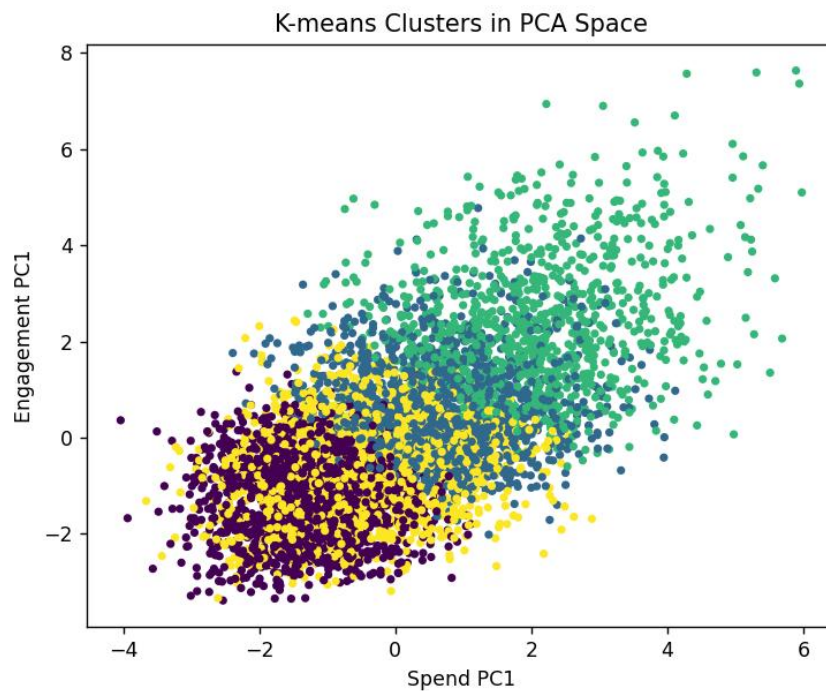


Figure 20 - K-means clusters in PCA space

Finally, we got detailed output for each customer cluster using mean values and Z-scores. While mean values are useful for understanding the absolute business characteristics of each cluster, the Z-scores highlight how each cluster performs relative to the overall customer average. For a detailed overview, see [Appendix: 3 – Cluster Scores](#).

4.1 Clustering Segments

The main purpose of our clustering analysis is to add business value and get customer segment insights. From the clustering results, there following four customer segments can be identified.

Cluster 1: Highly engaged and high-value customers

Cluster 1 represents high-value and highly engaged customers. This group has the highest average order value, the most frequent purchases in the last 3 months. (average orders almost 5 for the last 3 months) It is also noticeable that for this group, it shows the tendency that customers place more orders, spending more time on the website and app than others. This pattern suggests that further investment in online experience and personalized digital marketing could generate more revenue.

Cluster 2: Inactive or disengaged customers

Cluster 2 is a special cluster, which consists of the lowest spending, minimal engagement across channels, long recency periods, and low loyalty points. While the satisfaction score is not the lowest

among all clusters, this cluster lack of interaction and purchasing activity suggests limited business value in the short term.

Overall, the clustering analysis successfully caught the main variances and thus separated into 4 clusters for customers. It demonstrate that customers differ not only in value and engagement, but also in how service quality impact their satisfaction. These information provide a strong foundation for targeted marketing, business focus as well as service improvements.

Cluster 3: At-risk customers with service issues

Cluster 3 represents an at-risk customer segment, who shows a clear sign of dissatisfaction. Compared to other clusters, this group is more valuable for the business to identify weaknesses in service quality and prevent potential churn. These customers have noticeably higher complaint counts, return rates, longer delivery times, and the lowest satisfaction score. It indicates that customers who experienced poor deliver service, often have higher return rates as well as lower satisfaction scores. Earlier correlations heatmap also proves the negative relationship between satisfaction score and delivery times as well as complained rate.

Cluster 4: Satisfied but low-engagement customers

Cluster 4 represents a customer group who are satisfied but low-activity customers. These customers place relatively few orders and show moderate engagement, but they have the highest satisfaction score among others, which we can also see that they often experience faster delivery time and high on-time delivery. This also suggests that delivery services are critical for increasing customers' satisfaction.

5. Prediction Goals

5.1 Regression

Target variable: future_3m_spend

Core predictors: loyalty_tier_encoded, loyalty_points, orders_last_3m, recency_days, discount_rate_mean

Customer spending is a key business metric that directly reflects company revenue and long-term customer value. Because of this, we chose future_3m_spend as the continuous target variable, as it represents a forward-looking measure of customer value and directly supports business objectives related to demand forecasting and customer value management.

To maintain interpretability and avoid unnecessary model complexity, our goal is to first apply a baseline linear regression using a small set of core predictors reflecting customer spend behaviour and engagement. This allows us to identify how different customer characteristics and behavioural factors influence expected future spending. The linear regression also works as a transparent benchmark model for deeper analysis later on.

Predictor variables are selected based both on business relevance and insights from the earlier PCA analysis. Only variables that are known at or before the time of purchase are included to avoid data leakage. The final core predictor set consists of loyalty tier, loyalty points, recent purchasing activity, recency, and average discount rate.

Additional predictors related to product preferences and service experience are intentionally excluded from the baseline model. Instead, these variables are evaluated through Lasso regression, as they represent distinct customer behaviour and increase model complexity. By applying Lasso regularization, it becomes possible to assess whether these additional predictors contribute to meaningful explanatory insights while maintaining model interpretability.

5.2 Classification

As for binary variable, the choice was churn in upcoming 90 days, demonstrating if the customer is likely to place an order in the near future. This might be one of the most important parameters, since it is directly connected to revenue stability, fluctuations as well as customer retention. By predicting churn, we can gain powerful insights into customer behaviour, which would allow us to introduce proactive interventions such as discounts, loyalty tier upgrades, special bonuses, personalised outreach and other. Moreover, retaining existing customers and increasing their loyalty, by prolonging customer journey and providing high quality experience, is often cheaper and more efficient rather than acquiring new customers. Overall, prediction of churn in the next 90 days could reduce revenue loss, improve customer lifetime value, and optimise general marketing spend, by not only identifying pivotal churn drivers, but also by allocating marketing spend to higher risk segments. As a prediction model for this parameter the best fitting for several reasons would be Decision Tree. First and foremost, it is easy for interpretation and understanding for all of the stakeholders, considering the fact that this model sets clear and strict rules, for example, if customer's loyalty tier is low and they have not signed for email marketing, the risk of churn is high. And, on the opposite, if the customer was frequently placing orders in the past periods, the churn risk is lower. Furthermore, on the technical part, a decision tree can handle both categorical and binary features well, thus there is no need for encoding.

The categorical variable region of the customer is another key feature for classification because regional differences, usually, might affect customer preferences, churn rates, and marketing campaign effectiveness. So, classification by region helps business tailor localised strategies such as pricing, promotions, or product mix. The core value of this would be to identify underserved areas in order to improve market penetration and optimise further regional campaigns. For this, we will use the Logistic Regression prediction model mostly because it will provide us with precise features that impact regional differences the most. After encoding the model is suitable for categorical targets, furthermore, it avoids overfitting, compared to the single tree, finally, resisting noise. Overall, region classification would significantly contribute to explanation of differences in churn, loyalty, or customer engagement, tailoring of localized campaigns as well as building guidelines for resource allocation, for instance, marketing budget by region.

5.3 Additional Supervised regression – *KNN with Customer Satisfaction Score*

An additional supervised regression model has been included to assess if it is possible to predict customer satisfaction based on service quality, engagement, and behavioural variables. By understanding this possibility of prediction, some preaction can be made in advance.

We applied K-Nearest Neighbors (KNN) regression model for this task. Because KNN predicts satisfaction by comparing customers with similar service experiences and behavioural patterns, which aligns well with the subjective and comparative nature of satisfaction. In addition, KNN relies on local similarity instead of strong global assumptions such as random forest or decision tree, thus it allows to check whether customers who experience similar delivery performance, engagement levels, and purchasing behaviour also report similar satisfaction scores.

6. Predictive Modelling

6.1 Regression

Based on the previously mentioned prediction goals, regression modelling was used to evaluate both explanatory power and predictive performance. The analysis progresses from a simple, interpretable baseline model to more advanced regularized and dimensionality-reduced approaches. This structure allows for a direct comparison of how well original customer-level variables explain future spending, how redundancy and multicollinearity affect performance, and whether latent components extracted through PCA provide additional predictive stability.

The objective of the regression analysis was to examine how well customer-level behavioral, engagement, and service-related variables can explain future customer spending, measured as *future_3m_spend*. In particular, the analysis aimed to identify which variables best represent customer value and whether the key drivers suggested by exploratory analysis and PCA can meaningfully predict future revenue.

```
Linear Regression R2: 0.3657824407971981
Linear Regression RMSE: 15.551787839862863

Linear Regression Coefficients:
      Feature    Coefficient
1  loyalty_points    4.434469
0  loyalty_tier_encoded  4.317090
2  orders_last_3m    3.689654
4  discount_rate_mean -1.417966
3  recency_days     -1.033442
0
count    6000.000000
mean      0.622002
std       0.428093
min       0.000000
25%       0.143000
50%       1.000000
75%       1.000000
max       1.000000
```

Figure 21 - Linear Regression results

A baseline linear regression model was first estimated using core customer variables related to loyalty, recent purchasing activity, recency, and discount usage. The baseline model achieved a moderate explanatory power ($R^2 \approx 0.37$), indicating that these variables capture some relevant information about future spending but leave a substantial proportion of variation unexplained.

Although the direction of coefficients aligned with business intuition, such as higher loyalty and recent activity being associated with higher future spending, the model is likely affected by multicollinearity and overlapping information among predictors, as previously identified in the correlation analysis.

```

Lasso Regression R2: 0.6226232754540109
Lasso Regression RMSE: 11.996339482495381

Lasso Regression Coefficients (non-zero):
      feature    coefficient
8    returns_rate   -8.359443
2    orders_last_3m    3.768947
0    loyalty_tier_encoded  3.619838
7    satisfaction_score  2.367096
1    loyalty_points    1.992965
3    recency_days     -1.987886
6    email_click_rate_num -1.418659
4    discount_rate_mean -1.174689
9    on_time_rate      0.552770
5    email_open_rate_num  0.483225
10   category_diversity  0.024407

Chosen alpha (LassoCV): 0.027766798035107093
Number of non-zero predictors: 11

```

Figure 22 - Lasso Regression results

Given the multi-dimensional nature of customer behavior observed in the PCA results, Lasso regression was applied as a natural next step. Lasso introduces regularization that penalizes model complexity and shrinks less relevant coefficients toward zero, thereby addressing multicollinearity and improving model stability. Compared to the baseline model, Lasso regression achieved a substantial improvement in predictive performance, with an R^2 of approximately 0.62 and a noticeably lower RMSE. This demonstrates that regularization and feature selection are crucial when modeling customer behavior using correlated predictors. The regularization strength in the Lasso model was selected using cross-validation, ensuring that the level of penalization balances bias and variance rather than being manually tuned. This approach improves generalizability and reduces the risk of overfitting, in line with best practices for predictive modelling.

The Lasso coefficients provide clear and interpretable insights into the key drivers of future spending. Strong negative effects were observed for return behavior, recency, and discount dependence. In particular, a high returns rate was associated with a large reduction in future spending, highlighting return behavior as a strong indicator of customer dissatisfaction or friction. Similarly, customers who had not purchased recently and customers who relied heavily on discounts were less likely to generate future revenue, suggesting price sensitivity rather than long-term loyalty.

On the positive side, recent purchasing activity emerged as one of the strongest drivers of future spending, as reflected by the positive coefficient for orders placed in the last three months. Loyalty tier and accumulated loyalty points also showed strong positive effects, confirming that loyalty program

engagement is associated with higher future customer value. Service experience further contributed to future spending: higher satisfaction scores and higher on-time delivery rates were linked to increased future spending, reinforcing the importance of reliable service quality.

Marketing engagement variables, such as email open and click rates, showed smaller but still meaningful effects, indicating that engagement alone is less influential than transactional and service-related behavior once other factors are controlled for. The negative effect observed for certain engagement variables, such as email click rate, suggests that higher interaction may reflect promotion-driven or price-sensitive behaviour rather than genuine purchase intent, once other behavioural factors are controlled for. Product category diversity had only a marginal impact, suggesting that breadth of product interest plays a limited role compared to core behavioral drivers.

To assess whether dimensionality reduction could further improve model stability or performance, regression was also performed on PCA-transformed components.

```
PCA Regression R2: 0.5825968969618116  
PCA Regression RMSE: 12.616503968126294  
Number of PCA components: 5
```

Figure 23 - PCA Regression results

While PCA regression achieved better performance than the baseline linear model, it underperformed relative to Lasso regression. This indicates that although PCA successfully reduces multicollinearity by constructing orthogonal components, it also obscures important behavioral signals by combining multiple effects into latent factors. In this context, preserving original, interpretable customer variables, while applying regularization, proved more effective for predicting future spending.

Model	R ²	RMSE
Linear Regression	0.366	15.55
Lasso Regression	0.623	12.00
PCA Regression	0.583	12.62

Overall, the regression results highlight that future customer spending is primarily driven by recent purchasing behavior, loyalty engagement, service quality, and return behavior. Lasso regression provides the best balance between predictive performance and interpretability, making it well-suited for business decision support. The findings suggest that efforts to increase customer value should focus on improving service reliability, reducing returns, encouraging repeat purchases, and strengthening loyalty engagement, rather than relying solely on promotional discounts.



Figure 24 - Actual vs Predicted Future Spend

6.2 Classification

Starting with the first model of Decision Tree, we will use the churn in 90 days parameter as our target and other customer features as predicting factors. To create a cleaner dataset, we first delete customer id parameter, since identifiers do not contain any behavioural information. Next, we delete the high correlated monetary features like total spent, max and median order value, proceeding with only average order value. We also delete the future spending in 3 months column due to the fact that this data will not be proven when predicting churn, thus may impact our model's accuracy. We keep only the orders per last three months for the subset, since it captures both recency and frequency, and category diversity column, describing the idea clearly without any noise. Last, but not least, we exclude rows where email interactions are not relatable, because it is not a numerical feature in our dataset. At the end, the subset contains strong customer behavioural predictors also in terms of engagement, loyalty, operational experience, and segmentation in demographics. The next important step is to encode non-numeric features, namely, region and gender, to allow this model to work. For this reason, new categories were introduced: `new_categories = {1: 'North', 2: 'East', 3: 'South', 4: 'West'}` as well as `new_categories = {1: 'F', 2: 'M', 3: 'O'}`. Further, we divide our data into two parts, with 80% being the training and 20% test data. We proceed to creating a shallow tree, which is good for interpretability, not

performance, since it has low variance and only one best split, separating churners and non-churners. The visualisation of the tree illustrates which variable was used for the first split and the class distribution on each leaf. In evaluation, we compare training and test data in terms of accuracy, sensitivity, and specificity to detect possible overfitting. Finally, cross-validation trains 5 different trees on different folds and reports accuracy, giving a more stable estimate of performance. Accuracy scores of each fold in order: ['0.638239', '0.621733', '0.629986', '0.647383', '0.626722'].

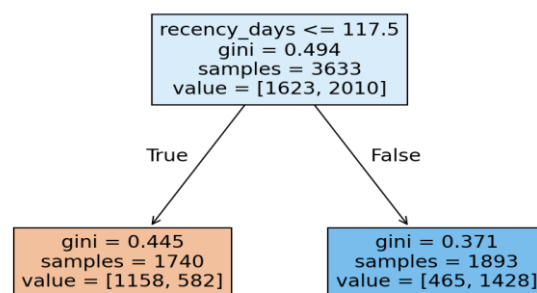


Figure 25 - Decision Tree, depth = 1

The decision tree model demonstrates that recency of last order is by far the strongest single predictor of customer churn. The tree splits customers at 117,5 days since last purchase, in other words almost four months, creating two distinct behavioural groups. In this way, customers who have placed an order within those last months show a much lower churn tendency, with the majority remaining prospective clients, while those who have been inactive for more than the indicated period have a significantly higher churn rate. This simple rule happens to highlight a powerful business insight, indicating customer inactivity as a strong warning signal for churn, making recency a highly actionable metric for retention strategies. Although the model is intentionally shallow and therefore easy to interpret, still, it provides a clear, operationalizable threshold that can be used to implement some of the marketing instruments, for example, to introduce timely interventions such as personalized offers, reminders, or loyalty incentives aimed at re-engaging customers before they leave completely.

The described decision tree predictive model offers rather high interpretability but moderate performance, which is exactly what is expected from a shallow model like this one. Since the tree uses only one split, its logic is extremely easy to understand and communicate, so this kind of transparency is valuable for business stakeholders and other people of interest. However, this simplicity comes at the cost of predictive power of the model. The cross-validation accuracy of roughly 63% illustrates that while recency is a meaningful churn signal, it cannot capture the full complexity of customer behaviour on its own, signifying that there are more parameters, influencing the probability of customer's churn.

All in all, the presented model is highly interpretable but only moderately accurate. In addition, considering everything said above, if we increase the max depth of the tree to 3, we can see that the next features, impacting the churn variable, would be return rate and loyalty points.

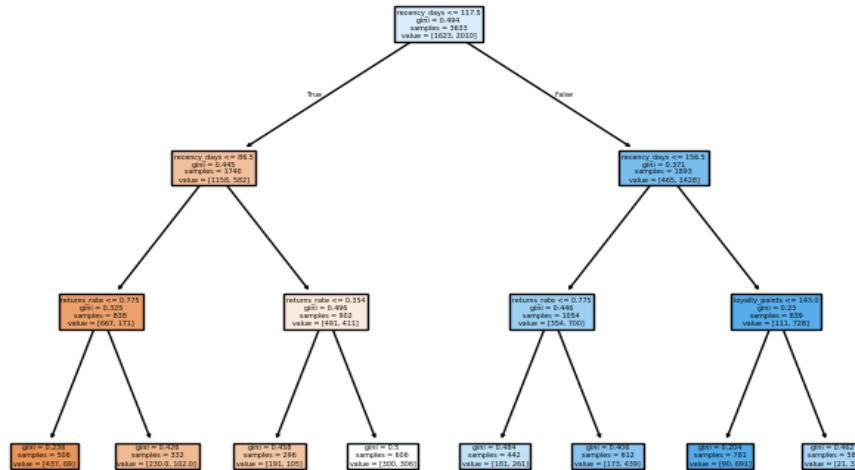


Figure 26 - Decision Tree, depth = 3

For the next categorical variable “Region”, the Logistic Regression predictive model was chosen. The data preparation follows the same pattern as in the previous Decision Tree model, only additionally excluding churn parameter, for the same reason we have deleted the future spendings. Logistic regression automatically switches to multinomial mode when the target has more than two classes. As a target parameter region North was chosen. Furthermore, ValueError occurred: Input X contains NaN. Even though the subset was already excluding the NR (Not Relatable) rows in email interactions, some predictors may contain empty strings, spaces or values failed conversion to numeric ones. To resolve this issue, we convert all numeric-like columns and drop rows with NaN. To interpret the results, we build a new data frame of coefficients, illustrating the effect of a certain feature. Positive coefficient increases the probability of the next customer ordering from that certain region, while negative – decreases. By doing this, we can clearly see which features differentiated region North and which behaviors are region-specific.

	coeff
age	-0.000266
email_opt_in	0.011190
loyalty_points	0.000536
loyalty_tier_encoded	0.107761
avg_order_value	-0.003963
pct_orders_discounted	-0.005663
recency_days	-0.000193
days_since_first_order	0.000015
orders_last_3m	-0.051793
distinct_products	-0.017659
category_diversity	0.012177
returns_rate	0.195607
complaint_count	0.150474
satisfaction_score	0.036687
email_open_rate	-0.004971
email_click_rate	0.047819
website_sessions_3m	-0.002260
app_sessions_3m	0.000290
avg_delivery_days	-0.080472
on_time_rate	0.165594
region_South	-6.307192
region_West	-6.130067
gender_M	0.013366
gender_O	0.586454

Figure 27 - Coefficients from Logistic Regression

The logistic regression model identifies several customer attributes that meaningfully distinguish North region customers from those in other parts. As expected, the strongest negative coefficients, in particular -6,30 and -6,13, belong to the dummy variables representing other regions, confirming that customers labeled with these categories are highly unlikely to order from the North region. Furthermore, the given model highlights several behavioural and operational characteristics associated with the chosen parameter. Following this, customers from the North tend to demonstrate higher return rates (0,19), better delivery on-time performance (0,16) also shown in negative coefficient on average delivery days (-0,08), with, however, higher number of complaints (0,15). On the other hand, not so many orders were placed from this region in the last three months, and the category diversity also has a negative coefficient value of -0,01. Moreover, other variables, such as age, recency, the amount of days since first order, product diversity, and app sessions, show coefficients close to zero, indicating minimal influence on regional classification. Finally, the northern customers are slightly more represented in higher loyalty tiers, and a notable positive coefficient appears in the “Other” gender category, suggesting demographic differences across various regions. To summarise, the logistic regression model reveals

that regional differences are driven mostly by operational experience, customer service, and loyalty characteristics, providing actionable insights for adjusting regional strategies and understanding geographic variation in customer behaviour.

The logistic regression model demonstrates solid and consistent performance in distinguishing North region customers from those in other parts, achieving 77% accuracy on the training set and 75% on the test set. The model is particularly strong at identifying customers who are not from the North, and as for the North region class itself, the model achieves high recall, successfully capturing the majority of true northern customers, though its lower precision indicates that some non-northern customers are incorrectly classified as such. This trade-off is typical in imbalanced classification settings and reflects a model that prioritises identifying as many customers from North as possible, even at the cost of some false positives. Overall, the results indicate a stable and generalisable model that provides a strong baseline for regional classification, offering useful predictive power while still maintaining interpretability and operational relevance.

6.3 Additional Supervised regression – *KNN with Customer Satisfaction Score*

In addition to the main regression task, a K-Nearest Neighbors (KNN) regression model was applied to assess whether customer satisfaction can be predicted from behavioural and service-related variables.

In this case, each customer is represented as a data point, which is presented by variables that measure service quality, customer engagement, and customer behaviors. Among variables, there are numeric data as well as Boolean type. While the numeric data are perfect to calculate distance, binary variables encoded as 0/1 were retained as valid indicators of similarity. Each data point we have chose many variables to define, which means that there are multiple dimensions to locate the datapoint. Although former performed PCA could help to reduce the dimensions, it will also reduces interpretability of supervised models and maximizes variance rather than predictive relevance. To preserve meaningful customer experience information and support clearer business interpretation, all original variables and thus dimensions are kept for KNN analysis.

KNN predicts satisfaction by measuring the distance between customers and identifying the most similar ones, with the final prediction calculated as the average satisfaction score of the nearest neighbors. Because KNN is distance-based, all variables were standardized to ensure that features with larger scales do not dominate the distance calculation. Model was performed by using both a train–test split and 5-fold cross-validation, as the two methods serve different but complementary purposes. The

train–test split provides an unbiased estimate of how well the model performs on completely unseen data, reflecting its expected real-world performance. On the held-out test set, the KNN regression model achieved an R^2 of approximately 0.62, indicating that it explains around 62% of the variance in customer satisfaction scores. The corresponding MAE (0.86) and RMSE (1.09) suggest that prediction errors are moderate and acceptable relative to the satisfaction score scale. In addition, 5-fold cross-validation was used to evaluate the stability and robustness of the model. By repeatedly training and validating the model on different subsets of the data, cross-validation reduces the risk that results depend on a single data split. The cross-validated R^2 of 0.61, together with similar MAE and RMSE values, is close to the test-set performance, indicating that the model generalizes well and is not overfitted.

The result of KNN regression as below:

Metris	Cross-Validation	Test Set
R^2	0.6117	0.6152
MAE	0.8754	0.8644
RMSE	1.1087	1.0880

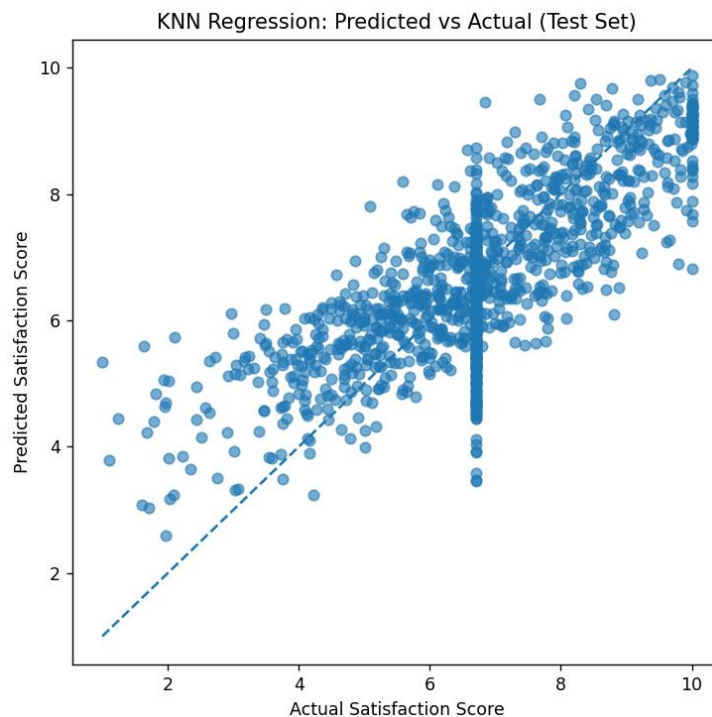


Figure 28 - KNN Regression

In addition, a predicted-versus-actual plot below was made to compare the model's predicted satisfaction scores with the true observed values on the test set. As below figure, each point represents an individual customer and the diagonal line indicating perfect prediction. Most of datapoints are around the diagonal line, which means that the prediction model captures key satisfaction patterns. There is a vertical concentration of points around the predicted score of 6 to 7.

This concentrated line of points is a characteristic behaviour of KNN regression and occurs because predictions are calculated as the average satisfaction score of the nearest neighbors. When many customers have similar profiles, their nearest neighbors overlap, resulting in similar predicted values.

Overall, the trained KNN regression model can be used to estimate customer satisfaction for new customer profiles by comparing them to similar past customers, enabling proactive service improvements and risk reduction.

7. Power BI Dashboard

To complement our dataset analysis and predictive modelling, a PowerBI dashboard was developed to provide an interactive and business-oriented view of the data and can be found attached as *Final_PowerBI_Dashboard.pbix*. The dashboard consists of four interconnected pages, each designed to support different analytical objectives and decision-making perspectives. Together, they translate complex analytical results into intuitive visual insights for non-technical stakeholders.

7.1 Customer Overview – Interactive Dashboard

The Customer Overview serves as the front page of the dashboard and provides a high-level summary of the customer base. Interactive slicers allow users to filter customers by gender, region, and loyalty tier, enabling flexible exploration across key segments.

Several visualizations are used to highlight core performance indicators, including average order value per customer, average spending by loyalty tier, and churn rate (90 days) by region. Radial gauges display the share of currently active customers and the overall average satisfaction score. This overview establishes a baseline understanding of customer value, engagement, and retention patterns, and sets the context for deeper analysis in the following dashboard pages.

Customer Overview – Interactive Dashboard

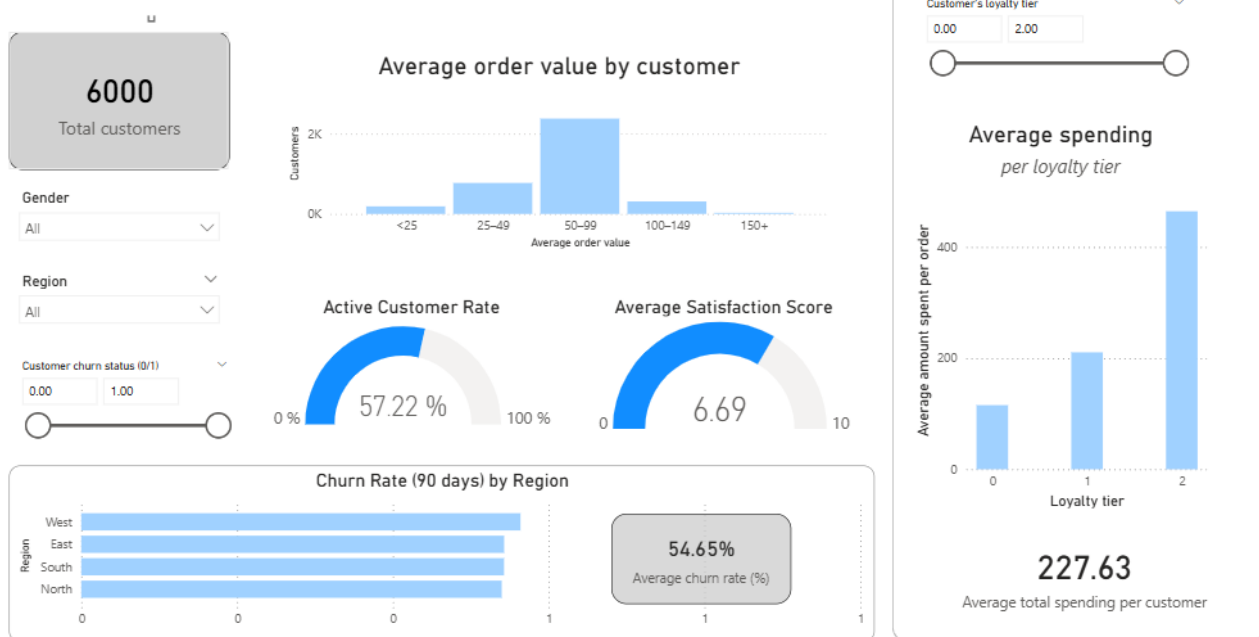


Figure 29 - PowerBI, Customer Overview

7.2 PCA Analysis and Customer Segmentation

In the second tab, we designed a dashboard that focused on customer segmentation. Values were provided by PCA-based K-means clustering. Such a visualization dashboard gives an even clearer view than the python results that the retail business has four customer segments, which highlight distinct business opportunities and risks.

At the top of the dashboard, an overview table summarizes the key characteristics of each customer segment by using average values. This allows stakeholders to quickly understand and compare cluster profiles at once. In addition, a slider for cluster labels is provided to enable focus analysis of individual cluster. Next, we have presented 4 PCA scatter plots below to visualize how clusters are distributed across the main behavioural dimensions, which were identified by PCA. Last but not least, an important KPI, satisfaction score bar chart, and an overview donut chart of cluster sizes, are included. The satisfaction score bar chart highlighted the customer satisfaction among cluster groups, while the overview of cluster sizes highlights both customer perception and the relative importance of each segment within the overall customer base.

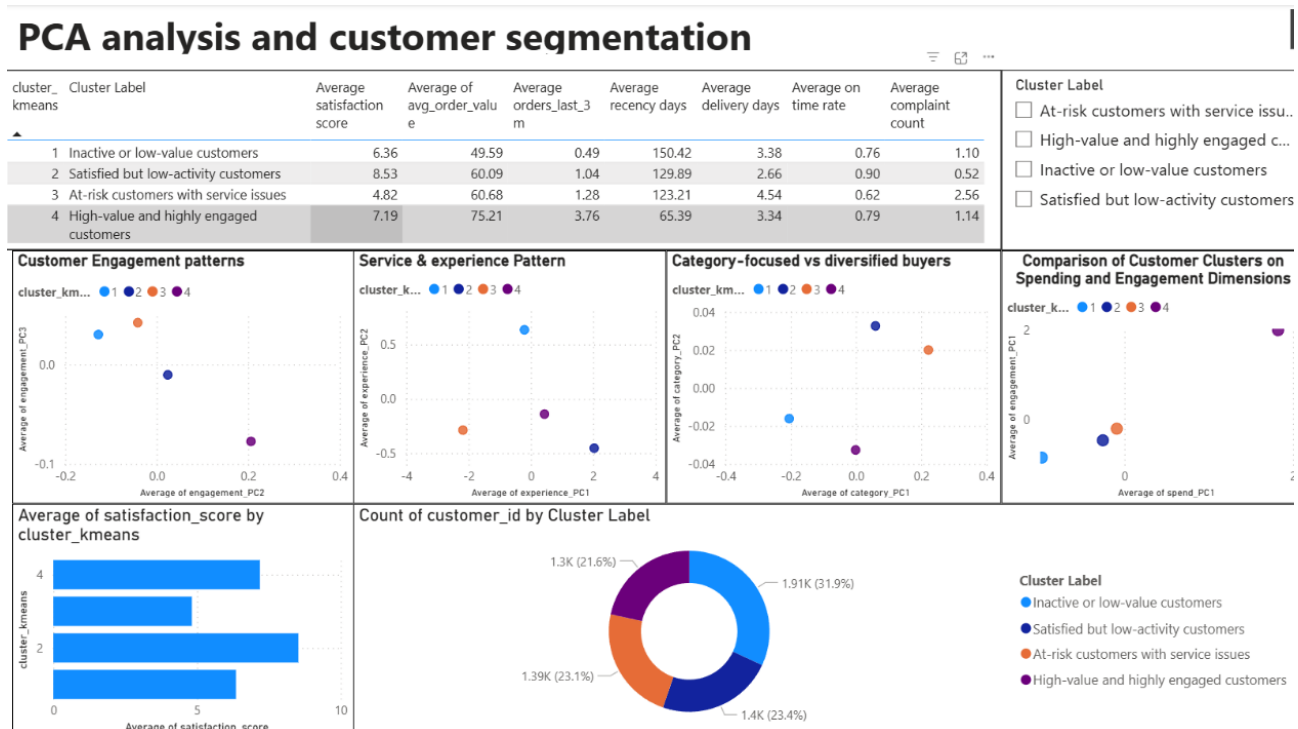


Figure 30 - PowerBI, PCA and Segmentation

7.3 Regression

The regression dashboard centers on the Lasso regression model, which demonstrated the strongest predictive performance. A detailed table combines customer-level information such as customer ID, average predicted future spend, historical spending, loyalty tier, cluster membership, and churn status. This allows direct comparison between predicted and observed behaviour.

A histogram of out-of-sample prediction errors (residuals) is included to assess model accuracy and stability. Additional visuals display average predicted spend by loyalty tier and the overall distribution of predicted future spending using binned column charts. These insights help stakeholders understand expected revenue patterns across customer segments and evaluate the financial impact of different loyalty and engagement strategies.

Regression

customer_id	Avg Predicted Spend (3 months)	Avg Historical Spend (3 months)	loyalty_tier_encoded	cluster_hierarchical	churn_proba_fullfit
1117	71.78	1,030.64	2.00	1.00	0.01
5674	71.25	978.93	2.00	1.00	0.01
3266	69.76	954.90	2.00	1.00	0.02
253	69.57	787.85	2.00	1.00	0.01
1958	67.08	889.14	2.00	1.00	0.01
2472	65.81	767.34	2.00	1.00	0.01
Total	17.95	109.95			

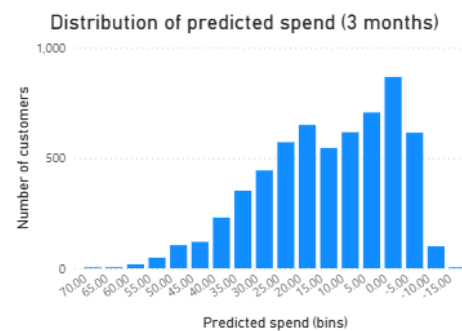
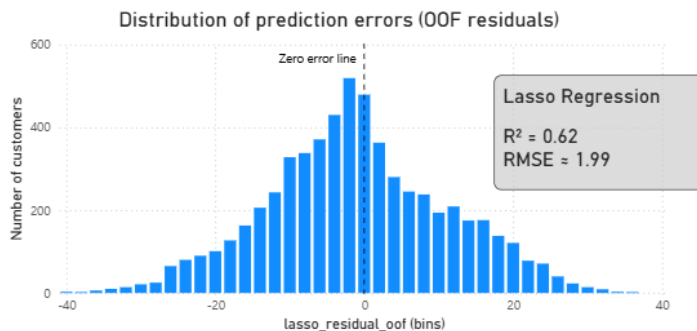
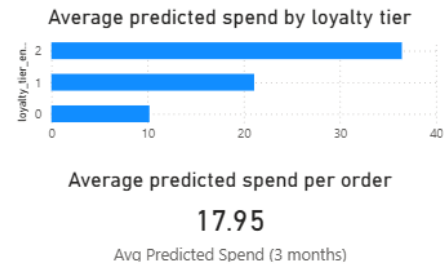


Figure 31 - PowerBI, Regression

7.4 Risk and Experience prediction

The final dashboard page focuses on customer risk, churn, and service experience. The first bar chart represents the average probability for customer's churn in each cluster, indicating also the total number of complaints as well as average return rate and days since last order, when hovering over a certain segment. The further scatter plot examines the relationship between number of delivery days on average and customers' satisfaction score. The trend line, which is the best-fit linear regression line drawn through all the data points, indicates a negative relationship between delivery performance and customer satisfaction. As it was mentioned earlier, slow delivery is harming customer experience, leading not only to lower satisfaction score, but also higher possibility of churn in the near future. The following tooltips, namely, cluster label and customer id work as identification, sum of complaints and orders per last three months as well as average number of delivery days and return rate explain what may influence the satisfaction score in general. The two card visuals represent the accuracy of KNN predictive model, using mean absolute error and root mean squared error. MAE equal 0,88 is as a moderate error, considering the scale from 1 to 10 used for this variable, meaning the model misses less by one point, which is acceptable for customer behaviour. On the other hand, RMSE of 1,11 means that when the model is wrong, the errors tend to be slightly larger than the average, however, it does not destabilise the model. Overall, the model is reliable for spotting low-satisfaction clusters, comparing

regions, identifying operational drivers and supporting customer experience strategy on the bigger scale. The simple line chart with average values for complaint count, return rate and link clicked rate, demonstrates the correlation between days since last order and churn probability, clearly showing the positive connection, confirming that recency is one of the strongest behavioural indicators of churn. Thus, customers who have not purchased for at least four months represent a high-risk segment and should be prioritized for re-engagement marketing campaigns. Finally, the slicers enable easy access for data by region, cluster and loyalty tier.

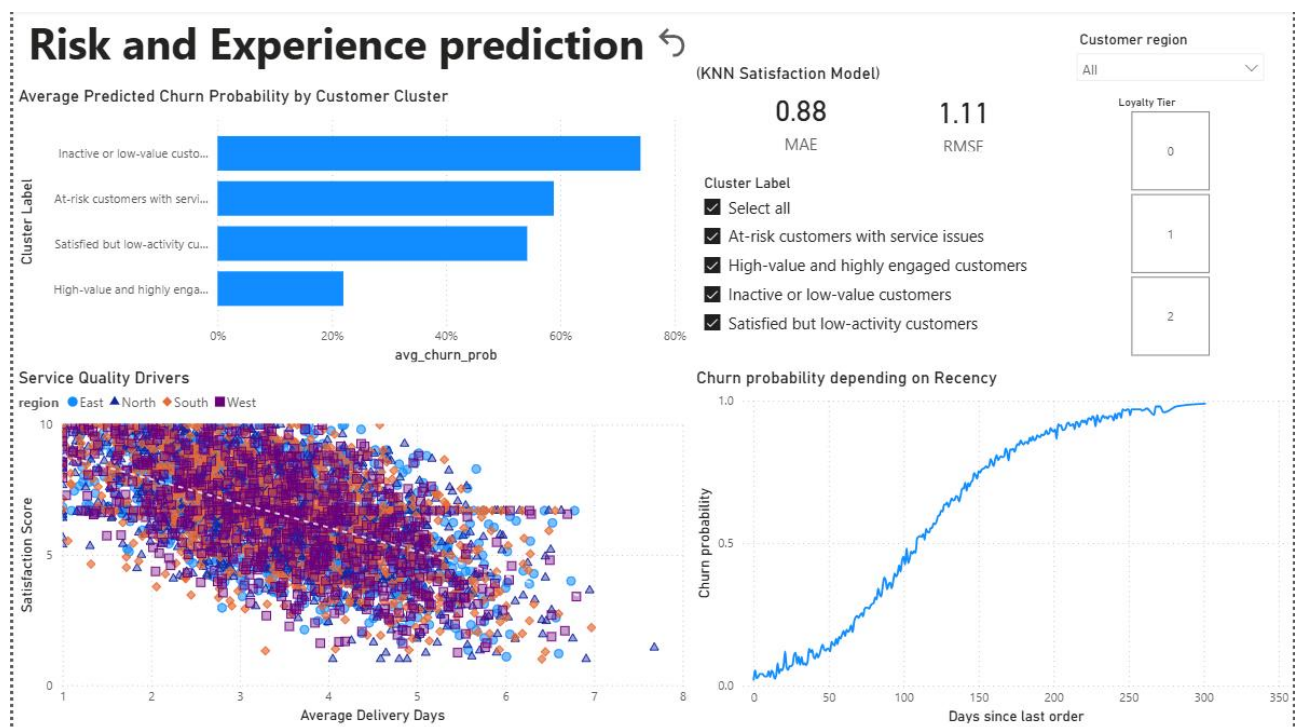


Figure 32 - PowerBI, Risk and Experience prediction

8. Conclusion

Overall, our data analysis demonstrates that customer behavior in this particular dataset is shaped by a combination of spending patterns, engagement, service quality, and product preferences rather than by a single dominant factor. Through careful data preparation, the dataset was transformed into a reliable analytical foundation, allowing meaningful patterns to emerge without being distorted by missing values, noise, or extreme outliers.

Exploratory analysis revealed several economically intuitive relationships, particularly between operational performance and customer experience. For instance, the strong association between on-time delivery and customer satisfaction highlights that logistics performance is not merely an operational concern, but a key driver of perceived value and loyalty. Such findings suggest clear

managerial actions, including closer monitoring of delivery partners and targeted improvements in the customer journey.

Dimensionality reduction and segmentation further confirmed that customer behavior is inherently multi-dimensional. While a single global PCA failed to capture behavior effectively, grouped PCA provided a clearer structure by separating spending, engagement, service experience, and product preference. Clustering based on these components revealed four distinct and interpretable customer segments, ranging from highly engaged, high-value customers to at-risk customers experiencing service issues. These segments provide a practical basis for differentiated marketing, retention, and service strategies.

Predictive modeling reinforced these insights. Regression analysis showed that future customer spending can be predicted with reasonable accuracy, with Lasso regression outperforming both baseline linear regression and PCA-based regression. The results indicate that recent purchasing behavior, loyalty status, satisfaction, and return behavior are the most influential drivers of future revenue. While the model does not capture all sources of uncertainty, it offers a strong balance between predictive performance and interpretability, making it suitable for business decision support rather than purely technical optimization.

Classification and additional predictive models further emphasized the importance of timely intervention. The churn analysis suggests that there is a limited window after a customer's last purchase in which proactive engagement can significantly reduce attrition. Similarly, the KNN regression results show that customer satisfaction can be reasonably predicted using observable service and engagement variables, enabling early identification of customers at risk of dissatisfaction and allowing for preventive service improvements.

Finally, the integration of analytical results into a PowerBI dashboard ensures that insights are accessible beyond the modeling environment. By combining raw KPIs, segmentation results, and model predictions into a single, interpretable dataset, the dashboard bridges the gap between advanced analytics and practical business use. Overall, the project demonstrates how structured data preparation, exploratory analysis, and predictive modeling together can support informed, data-driven decision-making across marketing, operations, and customer relationship management.

The analytical approach was guided by *Data Mining for Business Analytics* (Shmueli et al.), emphasizing business relevance, model interpretability, and appropriate use of predictive techniques. Likewise, principles from *Learning SQL* (Beaulieu) informed the construction of efficient and reliable queries, ensuring a solid foundation for subsequent data analysis.

Literature

Beaulieu, A. (2009): *Learning SQL (5th ed.)*. O'Reilly Media.

Shmueli, G., (2017): *Data Mining for Business Analytics*. John Wiley & Sons, Inc.

Attachments

SQL tables: *Tables_DataScienceExam.sql*

SQL queries: *Queries_DataScienceExam.sql*

Python file: *Data_Science_Exam.py*

PowerBI: *Final_PowerBI_Dashboard.pbix*

List of Figures

Figure 1 - Database overview in SQL.....	5
Figure 2 - Results of OOB and CV approaches for validation.....	8
Figure 3 - Total spent: Before vs After Imputation	9
Figure 4 - Total spent: KNN imputation	10
Figure 5 - Distribution of values by given parameters	11
Figure 6 - Correlation Heatmap.....	12
Figure 7 - PCA performed on all variables	13
Figure 8 - PCA performed on all variables, continued	14
Figure 9 - Customer Spend Behaviour, PCA results	15
Figure 10 - Customer Spend Behaviour, Variance.....	15
Figure 11 - Customer Interaction and Engagement, PCA results	16
Figure 12 - Customer Interaction and Engagement, Variance	16
Figure 13 - Service and Experience, PCA detailed	17
Figure 14 - Service and Experience, Variance.....	17

Figure 15 - Product Preference, PCA results	18
Figure 16 - Product Preference, Variance	18
Figure 17 - K-means clustering.....	20
Figure 18 - Clustering Dendogram.....	20
Figure 19 - Comparison of K-means and hierarchical clusters	21
Figure 20 - K-means clusters in PCA space	22
Figure 21 - Linear Regression results	26
Figure 22 - Lasso Regression results	27
Figure 23 - PCA Regression results	28
Figure 24 - Actual vs Predicted Future Spend.....	29
Figure 25 - Decision Tree, depth = 1	30
Figure 26 - Decision Tree, depth = 3.....	31
Figure 27 - Coefficients from Logistic Regression.....	32
Figure 28 - KNN Regression	34
Figure 29 - PowerBI, Customer Overview.....	36
Figure 30 - PowerBI, PCA and Segmentation	37
Figure 31 - PowerBI, Regression	38
Figure 32 - PowerBI, Risk and Experience prediction	39

Appendix

1. Tables

1.1 customers

We chose to limit a customer's birth year between 1900 and 2015, to avoid nonsense birth years. It is highly unlikely that a person born before 1900 (that would make them 125 years old) and after 2015 (someone younger than 10 years old) would put an order on a shopping site. We also chose to put a CHECK constraint on region, which is fixed to 'North', 'South', 'East' or 'West'. With the schema we were given, we figured it's unlikely that a region would be anything other than one of those four options. Likewise, we put checks on gender and loyalty tier, since the schema description stated that these values were limited to a few specific options.

1.2 products

The products table was the simplest one to make, since it only consists of a product ID together with its price, brand and category. The only check constraint we did is on the price ≥ 0 , to make sure the numeric value is always positive, but with the possibility of a product also being free in case of a discount like 'buy 3 and get 1 for free'.

1.3 orders

We know for a fact that each order belongs to a customer, so while we have order_id as primary key, we put customer_id as a foreign key in our orders table to correctly map each order with a customer.

The same way as in our 'products' table above, we put a check constraint on the total_amount, to make sure there are no negative totals, but still with the possibility of an order breaking even, for example in case of a faulty item that needs to be replaced. In this specific case, an order could have a total amount of 0. We were also told that the payment method for each order can be 'Card', 'PayPal', 'Transfer' or 'Other', so naturally we put a check constraint here as well.

1.4 order_items

We made the choice of adding one composite primary key in order_items from order_id and product_id. The composite key ensures that each order line is unique, and that a given product can only appear once within an order. If a customer for example buys 3 of the same product, it gets stored as quantity = 3 in the same row, instead of in duplicate rows. At the same time, it still allows the same product to appear across many different orders.

The check constraints we put in this table are by the columns `discount_pct` and `quantity`. A discount percentage between 0 and 1 enforces that discounts are valid proportions between 0% and 100%. This prevents impossible values like negative discounts or discounts above 100%, which would distort pricing/revenue analysis. Furthermore, a `quantity > 0` ensures that an order line represents an actual purchase quantity. Quantity at 0 would create meaningless lines and mess with the comprehension of the units sold.

1.5 order_returns

We considered renaming the table `returns` to `order_returns` to avoid confusion with the SQL keywords `RETURN` and `RETURNS`, which are reserved in SQL Server. However, renaming the table affects data loading, as the provided *exam_data.sql* file refers to the table name as just 'returns'. So without a modification to the file SSMS would turn back error when trying to insert values into the table.

To maintain consistency, we chose to rename the table to `order_returns` and update the `INSERT` statements accordingly. Given the limited size of the dataset, this was manageable and allowed us to use a clearer and more descriptive table name consistently throughout the schema.

An alternative solution would have been to keep the original table name and rely on square brackets when querying the table, but for this project we prioritised clarity in the schema design.

1.6 interactions

This table is more complex because several attributes are channel-specific. The fields 'opened' and 'clicked' are only relevant for the Email channel, while 'session_minutes' only applies to Web and App interactions. Therefore, these fields are allowed to be `NULL` when they are not applicable, for example when a customer didn't receive an email and therefore could not have opened and clicked it.

We defined the 'complaint' field as a `NOT NULL`able boolean with a default value of 0, since the dataset always explicitly records whether an interaction is a complaint or not. Support interactions may therefore have a value of 0 or 1, while interactions from other channels with no interactions in the Support channel will return the default value 0, indicating that no complaint occurred.

1.7 deliveries

We used booleans for `on_time`, indicating whether a delivery was completed on time (yes/no). We did not set a default value for this field like we did in some of the previous booleans, because deliveries are simply not always on time. Since both outcomes seem to occur in the dataset, the value needs to be clearly stated for each delivery, so it's always clear whether a delivery was on time or not.

We considered introducing a separate primary key (`delivery_id`) to uniquely identify delivery records. However, an analysis of the relationship between orders and deliveries showed a one-to-one relationship, where each delivery corresponds to exactly one order. Consequently, `order_id` was used as the primary key in the deliveries table, serving both as a unique identifier and as a foreign key to the orders table. At the same time we aim for data minimization, by leaving out unnecessary complexity that would not add analytical value. While introducing additional keys could be useful in other scenarios, we chose a simpler structure here to keep the model easy to understand and reduce the risk of errors or unintended ripple effects across the data.

2. Individual Questions

Ruyi Zhong

1. What is the age distribution of customers within each loyalty tier?(Age groups: 0–18, 19–35, 36–55, 56–70, 70+). The output helps to identify the customer segment for each loyal tiers, thus identify the business main target age group.
2. What is the gender distribution within each loyalty tier? The output is useful to understand demographic differences between high- and low-value customer groups.
3. What is the average quantity of items per order across all customers? The output shows typical basket size thus it is useful for inventory planning and pricing strategy.
4. For each loyalty tier, what is the most commonly purchased order quantity? The output reveals behavioural differences between tiers (e.g., premium customers buying in larger volumes).
5. List all products with their profit per unit (approx. net revenue per unit). Identifies high-margin products that should be promoted or prioritized.
6. List and rank all products' total profit. Shows the overall financial impact of each product, combining margin and sales volume.
7. What percentage of customers have submitted at least one complaint, and how many total complaints were recorded? It provides an overview to see the feedback from customers. if there are high presentage of complaints among customers, thus we will need to dive into details.
8. What percentage of deliveries were not completed on time? A core operational KPI affecting customer satisfaction and repeat purchase rates.

Sasja Stein

- Loyalty Tier Insights
 1. Average order value by loyalty tier (Premium / Plus / Basic):
 - Is there a connection between loyalty tier and how much people spend?
- Product Insights
 2. Top 10 products sold by quantity
 3. Which product is the most popular (most sold) within each loyalty tier?
- Payment Method Insights
 4. How much revenue do we get from each payment method?
 5. Within each payment method, how many orders come from opted-in vs not opted-in customers?
- Regional and Deliverance Insights
 6. Customers and orders by region:
 - Which region places the most orders?
 7. How many delayed deliveries are there in each region?
 8. Are complaints more common for late deliveries, and does it vary by region?

Svetlana Podolskaja

1. Is there a big time difference between order and shipping date by region of the customer? If so, it may come due to some technical issues that we need to tackle, for example, understocking in the warehouses, inefficient partnership with shipping companies, etc.
2. Did the customer place a new order after leaving a complaint through Support? Can be one of the KPI's for Customer Support department.
3. Is email channel effective among young customers (between 1997 and 2010 birth year)? Shows if this marketing tool is beneficial among generation Z, or should we proceed with other more accurate solutions.
4. How many orders have been placed on the same day as interaction with our promotion channels happened? After grouping, demonstrates the most efficient marketing channels.
5. Are there any products which are being returned more often than kept? If yes, we would probably need to examine them, especially, if this happens due to the product being Defective.
6. Do the orders with bigger difference between order and shipping date have more returns, where the reason wouldn't be Defect or Size? Could we possibly minimize the amount of returned orders by optimising the speed of shipping?

7. Are there any customers repeatedly ordering the same products in large amounts? A possibility to build stronger connections and increase customer loyalty by further customised suggestions and special offers.
8. Are there a lot of return due to Size? If this is the case in more then 50%, should we change or add the size table, or add a bigger button to double check size in order to minimise return cases in the future.

3. Cluster Scores

=== Cluster Profile (Mean Values) ===					=== Cluster Profile (Z-scores) ===				
cluster	avg_order_value	orders_last_3m	discount_rate_mean	recency_days \	cluster	avg_order_value	orders_last_3m	discount_rate_mean	recency_days
1	76.65	4.09	0.37	58.73	1	1.16	1.32	-0.93	-1.22
2	47.67	0.22	0.42	160.24	2	-1.20	-0.95	1.39	1.07
3	66.68	2.17	0.38	97.13	3	0.35	0.19	-0.46	-0.35
4	58.47	0.87	0.39	134.43	4	-0.32	-0.57	0.00	0.49
website_sessions_3m app_sessions_3m email_open_rate \					website_sessions_3m app_sessions_3m email_open_rate \				
cluster	website_sessions_3m	app_sessions_3m	email_open_rate	\	cluster	website_sessions_3m	app_sessions_3m	email_open_rate	\
1	3.02	2.69	0.64		1	1.37	1.37	1.20	
2	1.02	1.07	0.33		2	-0.90	-0.85	-1.05	
3	1.88	1.77	0.53		3	0.08	0.11	0.40	
4	1.32	1.24	0.40		4	-0.56	-0.62	-0.54	
email_click_rate loyalty_points satisfaction_score \					email_click_rate loyalty_points satisfaction_score \				
cluster	email_click_rate	loyalty_points	satisfaction_score	\	cluster	email_click_rate	loyalty_points	satisfaction_score	\
1	0.28	351.98	7.71		1	1.23	1.40	0.69	
2	0.05	34.63	5.64		2	-0.99	-0.88	-0.80	
3	0.19	154.10	5.49		3	0.36	-0.02	-0.91	
4	0.09	86.53	8.16		4	-0.60	-0.51	1.02	
complaint_count returns_rate avg_delivery_days on_time_rate \					complaint_count returns_rate avg_delivery_days on_time_rate \				
cluster	complaint_count	returns_rate	avg_delivery_days	on_time_rate \	cluster	complaint_count	returns_rate	avg_delivery_days	on_time_rate \
1	0.89	0.39	3.11	0.83	1	-0.56	-0.75	-0.52	0.55
2	1.71	0.89	3.87	0.70	2	0.57	1.00	0.59	-0.69
3	2.08	0.81	4.19	0.67	3	1.08	0.72	1.06	-0.98
4	0.50	0.33	2.69	0.89	4	-1.09	-0.96	-1.13	1.12
category_share_electronics category_share_grocery \					category_share_electronics category_share_grocery \				
cluster	category_share_electronics	category_share_grocery	\		cluster	category_share_electronics	category_share_grocery	\	
1	0.29	0.25			1	-0.00	0.00		
2	0.28	0.26			2	-1.22	1.22		
3	0.30	0.24			3	1.22	-1.22		
4	0.29	0.25			4	-0.00	0.00		
category_share_home					category_share_home				
cluster	category_share_home				cluster	category_share_home			
1	0.20				1	-0.5			
2	0.21				2	1.5			
3	0.20				3	-0.5			
4	0.20				4	-0.5			